Quiz 1 grade improvement

October 12, 2026

Problem 1. *Let $\{a_n\}$ be a sequence of positive rational numbers (i.e. $a_n > 0$) such that $\lim_{n\to\infty} a_n = m$.*

1. *(2.5 points) Prove that for n large enough $\dfrac{a_n}{2} < \dfrac{3m}{4}$.*
2. *(2.5 points) Prove that for n large enough $\dfrac{a_n}{2} < \dfrac{5m}{8}$.*
3. *(1 point) Plot the above two numbers on the real line to get their relative positions. What do you intuitively understand from the above two problems? Explain in words. (I am curious to know what everyone gets out of this, this is a subjective question, so there is no right or wrong answer.)*

Problem 2. *(2 points) Skim through the lecture notes and reflect on what you have learned in this course so far.*

Try to explain, in as many or as few words as you like, why you think we studied the topics we did, and how the different ideas connect to one another. Do you see a logical progression emerging from what we have learned?

Finally, make a list of the mathematical techniques you have learned in this course so far. (Mathematical reflection is an important exercise in learning math. It helps us see things from a bird eye's perspective and understand better.)